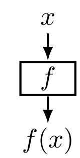
L1.1a Functions & Catalog (§1.1–1.2)

What you need to know

A FUNCTION, FOUR WAYS

Algebraic · **numerical** · **graphical** · **verbal** — one object, four views. f gives each x in D **exactly one** $f(x)$ in E; its graph is the set

$$\boxed{\{\,(x, f(x)) \mid x \in D\,\}}$$



"Think of f as a machine — the output must be defined."
D = every input R = every output **VLT**: no vertical line hits it twice.

DOMAIN & RANGE

$$\boxed{\text{even root's inside } \geqslant 0 \qquad \text{denominator } \neq 0}$$

D off the formula, R off the outputs. **Both notations:** $\{x \mid x \neq 0, 1\} = (-\infty, 0) \cup (0, 1) \cup (1, \infty)$
Sign probes, not a sign chart: sign each factor `@`-1, `@`0.5, `@`10, then read the whole.
"Intersection cuts out just the overlap of the two domain intervals. Union gives you any place that either of them goes."

The catalog — nine families

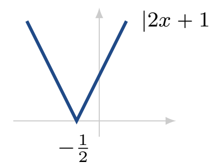
Family	Form	Domain / range	What marks it
Linear	$mx + b$	$D = R = (-\infty, \infty)$	$m = \frac{\Delta y}{\Delta x}$, constant everywhere
Polynomial	$a_n x^n + \cdots + a_0$	$D = (-\infty, \infty)$	smooth, $\leqslant n-1$ turns; factor with **Loh**
Power	x^n	even n: $R = [0, \infty)$; odd n: $R = (-\infty, \infty)$	x^6 is a *steeper* x^2, not a new shape
Root	$\sqrt[n]{x}$	even n: $D = [0, \infty)$; odd n: $D = (-\infty, \infty)$	**odd roots have negative domains**
Rational	$\dfrac{P(x)}{Q(x)}$	D: every x with $Q(x) \neq 0$	`Z@` · `VA@` · `HE@` · `HA`
Algebraic	$\sqrt{x^2 + 1}$	whatever survives every root and denominator	polynomials put through $+\ -\ \times\ /$ and roots
Trig	$\sin x,\ \cos x,\ \tan x$	$\sin, \cos$: $R = [-1, 1]$; $\tan$: $x \neq \frac{\pi}{2} + \pi k$	sin, tan odd; cos even; all periodic
Exponential	b^x, $b > 0$	$D = (-\infty, \infty)$, $R = (0, \infty)$	`HA y=0`; the variable is the **exponent**
Logarithmic	$\log_b x$	$D = (0, \infty)$, $R = (-\infty, \infty)$	`VA@0`, `Z@1`; "what exponent?"

Loh, never the quadratic formula: divide by the leading coefficient, then $m = -\frac{b}{2}$, $d = \sqrt{m^2 - c}$, roots $m \pm d$.
E.g. $3x^2 + 2x - 1 \to x^2 + \frac{2}{3}x - \frac{1}{3}$: $m = -\frac{1}{3}$, $d = \frac{2}{3}$, roots $\frac{1}{3}, -1$, so $(3x - 1)(x + 1)$.

`HA`, three cases: $\deg(\text{bot}) > \deg(\text{top}) \Rightarrow$ `HA y=0` · equal $\Rightarrow$ ratio of leading coefficients · $\deg(\text{bot}) < \deg(\text{top}) \Rightarrow$ long division, `SA`.

PIECEWISE & ABSOLUTE VALUE

$$\boxed{|x| = \begin{cases} x & x \geqslant 0 \\ -x & x < 0 \end{cases}}$$



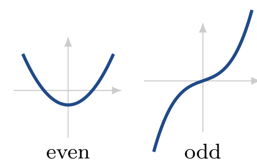
"chop and reflect at the vertex"

Four substitutions. $2x+1$ goes in for *each* x — both formulas **and** both restrictions, which are inequalities to solve.

$$|2x+1| = \begin{cases} 2x+1 & 2x+1 \geqslant 0 \;\Rightarrow\; x \geqslant -\frac{1}{2} \\ -(2x+1) & 2x+1 < 0 \;\Rightarrow\; x < -\frac{1}{2} \end{cases}$$

SYMMETRY

$$\boxed{f(-x) = f(x) \;\; \text{even} \qquad f(-x) = -f(x) \;\; \text{odd}}$$

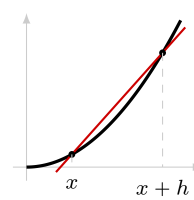


"symmetric = even, antisymmetric = odd"

Test the algebra, not the picture. Most functions are **neither** — $x - x^2$.

THE DIFFERENCE QUOTIENT

$$\boxed{m = \frac{\Delta y}{\Delta x} = \frac{f(x+h) - f(x)}{h}}$$



"average rate of change — the slope of the secant"

$x^2 + x \;\Rightarrow\; 2x + 1 + h$. A line keeps neither h nor x: $4x - 1 \Rightarrow 4$.

WATCH OUT

- ✗ ~~$\sqrt[3]{-8}$ is undefined~~ → **odd roots have negative domains — $\sqrt[3]{-8} = -2$**
- ✗ ~~$|2x+1|$ takes two substitutions~~ → **four — both formulas *and* both restrictions**
- ✗ ~~a cancelled factor comes back~~ → **that input is still gone — it is a hole, `HE@`, not a `VA`**
- ✗ ~~every function is even or odd~~ → **most are neither — $x - x^2$**
- ✗ ~~the difference quotient is the tangent slope~~ → **it is the *secant* slope — an average over a step h**
- ✗ ~~$\cup$ and $\cap$ both mean "combine"~~ → **$\cap$ keeps only the overlap; $\cup$ keeps anywhere either goes**
- ✗ ~~use the quadratic formula~~ → **Loh, always**